Teacher's Assessment Activity

DATA STRUCTURE AND ALGORITHMS

24EE05TH0505-2

5th Semester B.Tech. Session-2026-27

TA Report

RBU

RAMDEOBABA UNIVERSITY, NAGPUR

Formerly Shri Ramdeobaba College of Engineering & Management (RCOEM) Est. 1984

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Title : Airport Flight Management System

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1. Introduction :

Air transportation plays an important role in connecting cities and countries. Efficient airport management requires proper handling of flight schedules, passenger information, ticket bookings, seat availability, airport routes, and gate allocation. Managing these operations manually can be complex and time-consuming. Therefore, a computerized system is useful for organizing flight information and simplifying airport management activities.

AeroFlow ATOS (Airport Traffic & Operations Simulator) is a web-based Airport Flight Management System developed to demonstrate the practical implementation of Data Structures and Algorithms (DSA). The application provides features such as flight management, flight searching, ticket booking, seat selection, booking cancellation, passenger management, departure queues, and airport route exploration. Users can search for flights, select available seats, calculate simulated fares, and generate booking confirmations.

The project incorporates various data structures, including arrays, stacks, queues, linked lists, binary search trees, AVL trees, heaps, hash tables, and graphs. Searching and sorting algorithms help organize and retrieve flight information efficiently. Breadth-First Search (BFS) and Depth-First Search (DFS) are used to explore airport connections, while Dijkstra's algorithm can identify minimum-distance routes in a weighted airport network. Queues manage departure processing, and priority queues demonstrate priority-based flight handling.

The application is developed using modern web technologies such as React, TypeScript, and Vite, with browser-based storage for supported records. Its interactive interface helps users understand how different data structures and algorithms operate in practical scenarios. The project also provides an opportunity to study algorithmic complexity and improve programming and problem-solving skills.
The main objective of this project is to bridge the gap between theoretical DSA concepts and their practical applications through a realistic airport-management simulation. It demonstrates how efficient data organization and algorithm design can support flight management, booking operations, and airport resource allocation.

2. Problem Statement and Objectives

2.1 Problem Statement

Managing airport operations involves handling flight schedules, passenger details, ticket bookings, seat availability, flight routes, and gate allocation. Managing these activities separately can make information retrieval, booking management, and resource allocation complicated. A systematic application is required to organize flight information, simplify booking procedures, prevent duplicate seat assignments, and demonstrate efficient data processing. Therefore, the Airport Flight Management System (AeroFlow ATOS) is designed to simulate airport operations while demonstrating the practical application of Data Structures and Algorithms (DSA).

2.2 Objectives

The main objectives of this project are:

- **To develop a user-friendly web application for managing flight schedules and passenger information.**
- **To implement flight searching, ticket booking, seat selection, fare calculation, and booking cancellation.**
- **To use arrays, stacks, queues, linked lists, trees, heaps, hash tables, and graphs for efficient data organization.**
- **To implement searching and sorting algorithms for retrieving and organizing flight records.**
- **To demonstrate BFS, DFS, and Dijkstra's algorithm for exploring airport connections and finding routes.**
- **To manage departure queues, simulated flight priorities, and airport gate allocation.**
- **To maintain booking records and seat availability consistently within the application.**
- **To analyze algorithmic time complexity and understand the practical applications of DSA.**
- **To provide an interactive platform for demonstrating DSA concepts through a realistic airport-management simulation.**

3. System Requirements and Technologies Used

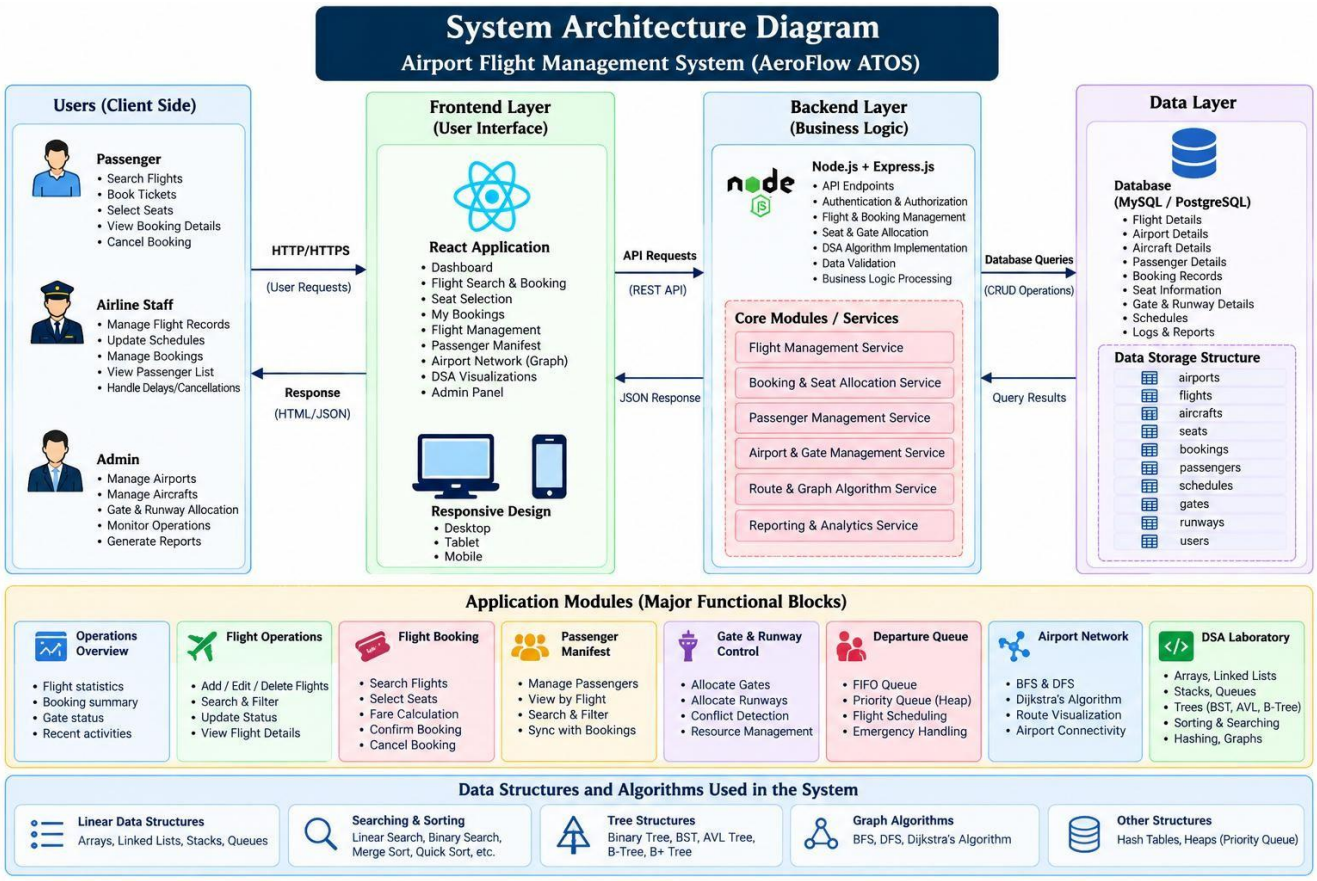
3.1 Hardware Requirements

A computer capable of running a modern web browser and a current Node.js environment is sufficient for the local demonstration. No dedicated hardware is required

3.2 Software and Tools

- **Frontend: React with TypeScript and Vite.**
- **Styling and icons: CSS and Lucide React.**
- **Visualizations: SVG-based graph and tree views.**
- **Persistence: Browser LocalStorage for supported application records.**
- **Testing: Vitest, plus manual browser verification.**
- **Version control and submission: Git and GitHub.**

4. Architecture Diagram



5. Implementation and Working Principle

5.1 Implementation

The major modules include:

- **Operations Overview: Displays a summary of flight operations, bookings, and other available statistics.**
- **Flight Management: Allows users to view, search, filter, sort, and manage flight records.**
- **Flight Booking: Enables users to search for flights, enter passenger details, select available seats, calculate simulated fares, confirm bookings, and cancel existing bookings.**
- **Passenger Management: Organizes passenger information and associates passengers with their respective bookings.**
- **Gate and Runway Control: Demonstrates resource allocation and detects scheduling conflicts.**
- **Departure Queue: Uses queue and priority queue concepts to simulate flight departure processing.**
- **Airport Network: Represents airport connections as a graph and demonstrates BFS, DFS, and Dijkstra's algorithm.**

5.2 Working Principle

Step 1: Accessing the Application
The user opens the web application in a browser and accesses the dashboard to view the available flight-management and booking features.

Step 2: Managing Flight Information
Flight records are displayed in the flight management module. Users can search for specific flights, apply filters, sort records, and view available flight details.

Step 3: Searching and Booking Flights
The user enters the departure airport, destination, travel date, number of passengers, and cabin preference. The application searches the available sample flight records and displays matching results.

Step 4: Selecting Seats and Confirming Bookings
The user selects a flight, enters passenger details, chooses an available seat, and views the simulated fare. After confirmation, the application generates a booking reference and updates the corresponding booking and seat-availability records.

Step 5: Managing Existing Bookings
Users can view their booking details and cancel eligible bookings. When a booking is cancelled, the application updates the booking status and releases the selected seat, subject to the implemented booking rules.

Step 6: Storing and Displaying Information
The application updates the relevant interface components after supported operations. Records stored using LocalStorage remain available across browser sessions on the same browser, subject to storage being retained.

6. Conclusion

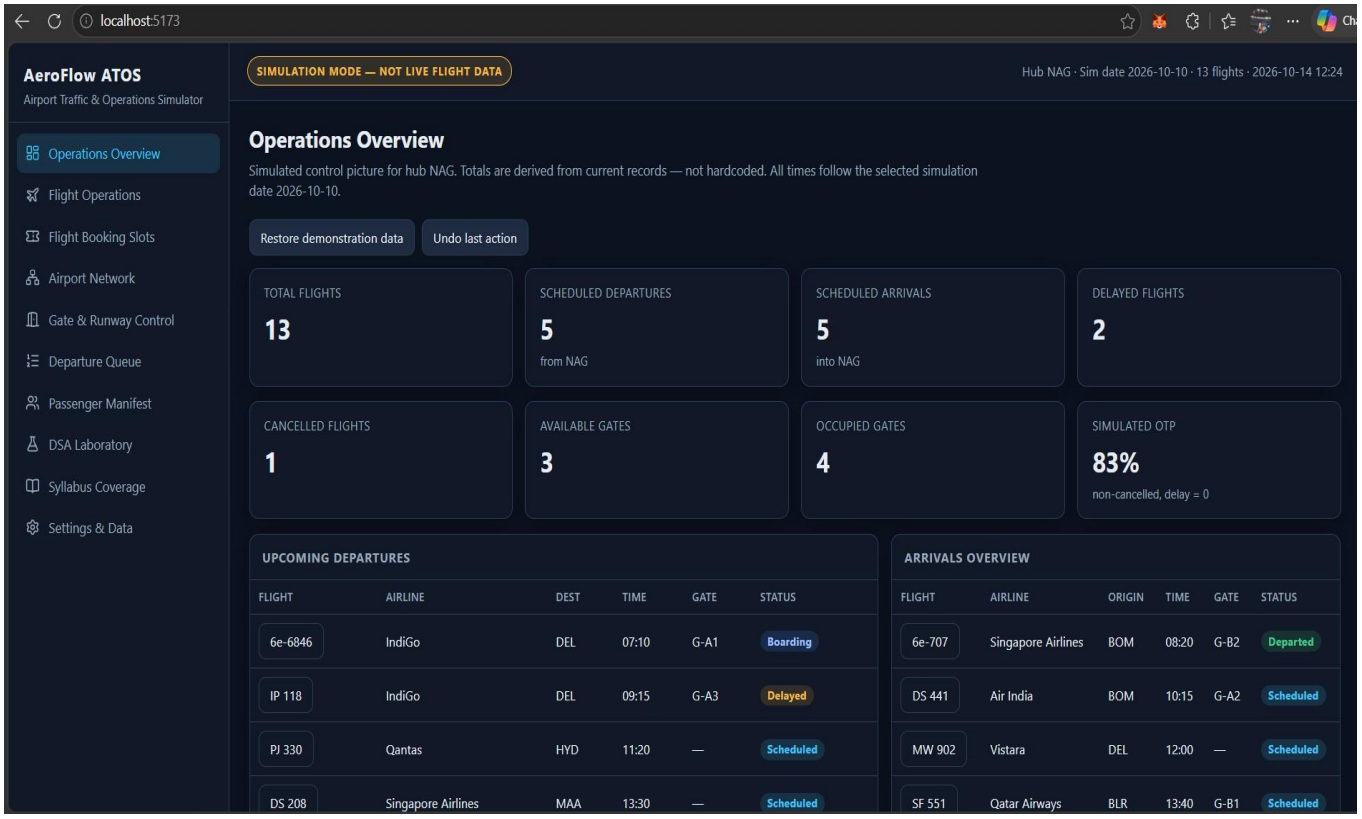
The Airport Flight Management System (AeroFlow ATOS) demonstrates the practical application of Data Structures and Algorithms (DSA) in airport management through a web-based simulation. The project provides a structured approach to managing flight information, searching flights, booking tickets, selecting seats, maintaining passenger records, and exploring airport connections.

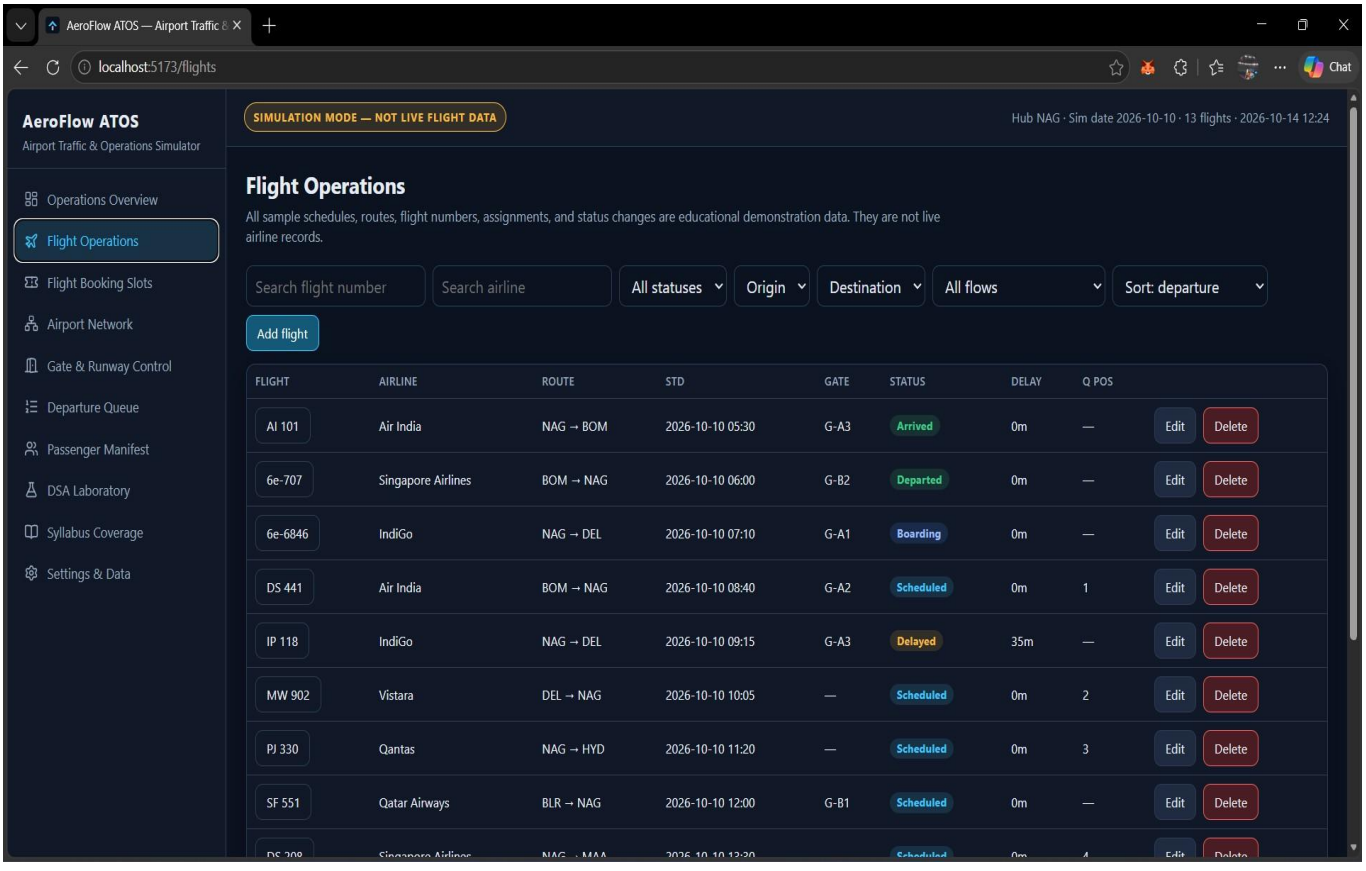
Various data structures, including arrays, stacks, queues, linked lists, trees, heaps, hash tables, and graphs, illustrate how information can be organized and processed efficiently. Searching and sorting techniques help retrieve flight records, while BFS, DFS, and Dijkstra's algorithm demonstrate airport network traversal and route calculation. The project also highlights the importance of algorithmic complexity and efficient data management in software development.

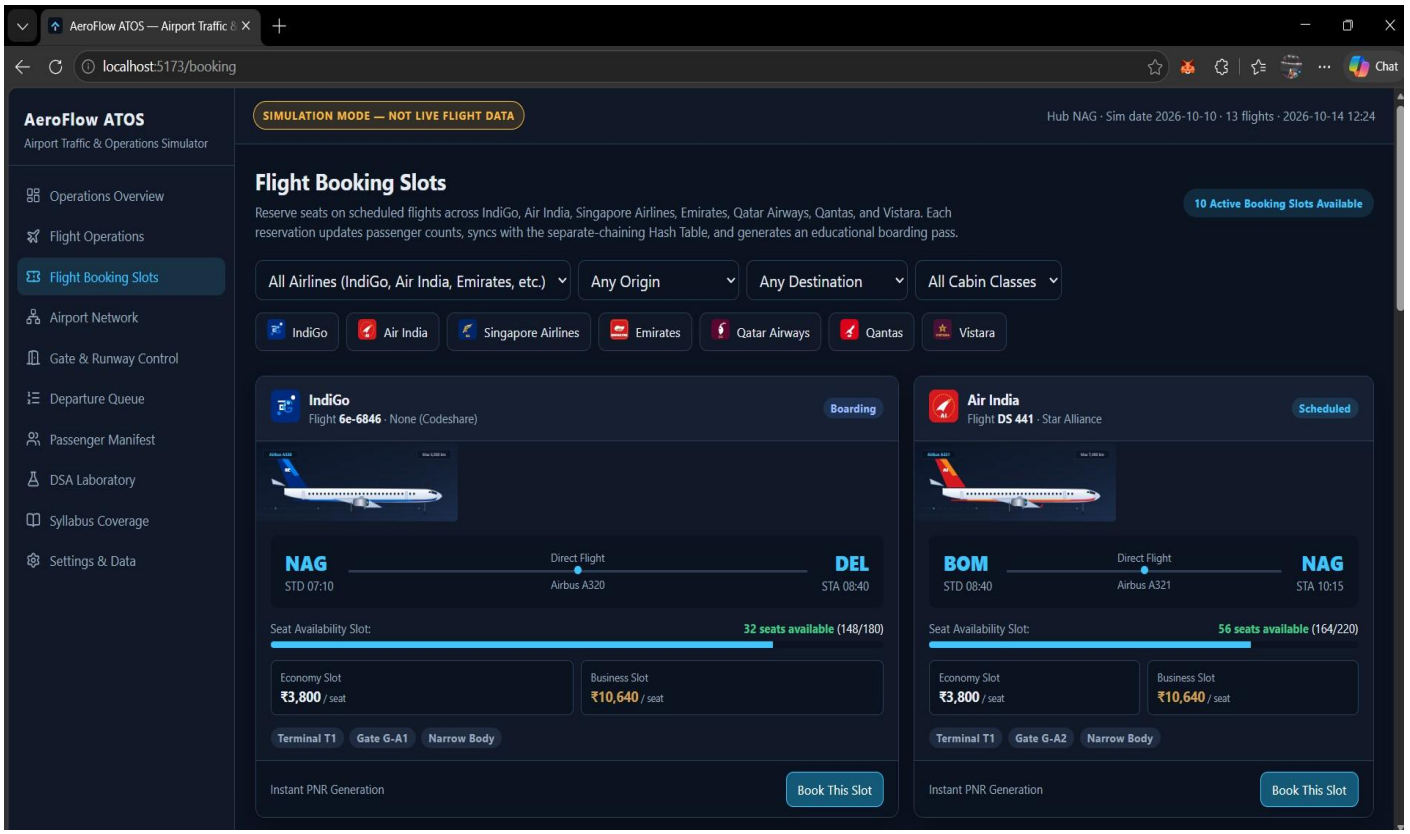
Developing this project provides practical experience in frontend development, application logic, data handling, problem-solving, and teamwork. It helps bridge the gap between theoretical DSA concepts and their practical implementation through an interactive application.

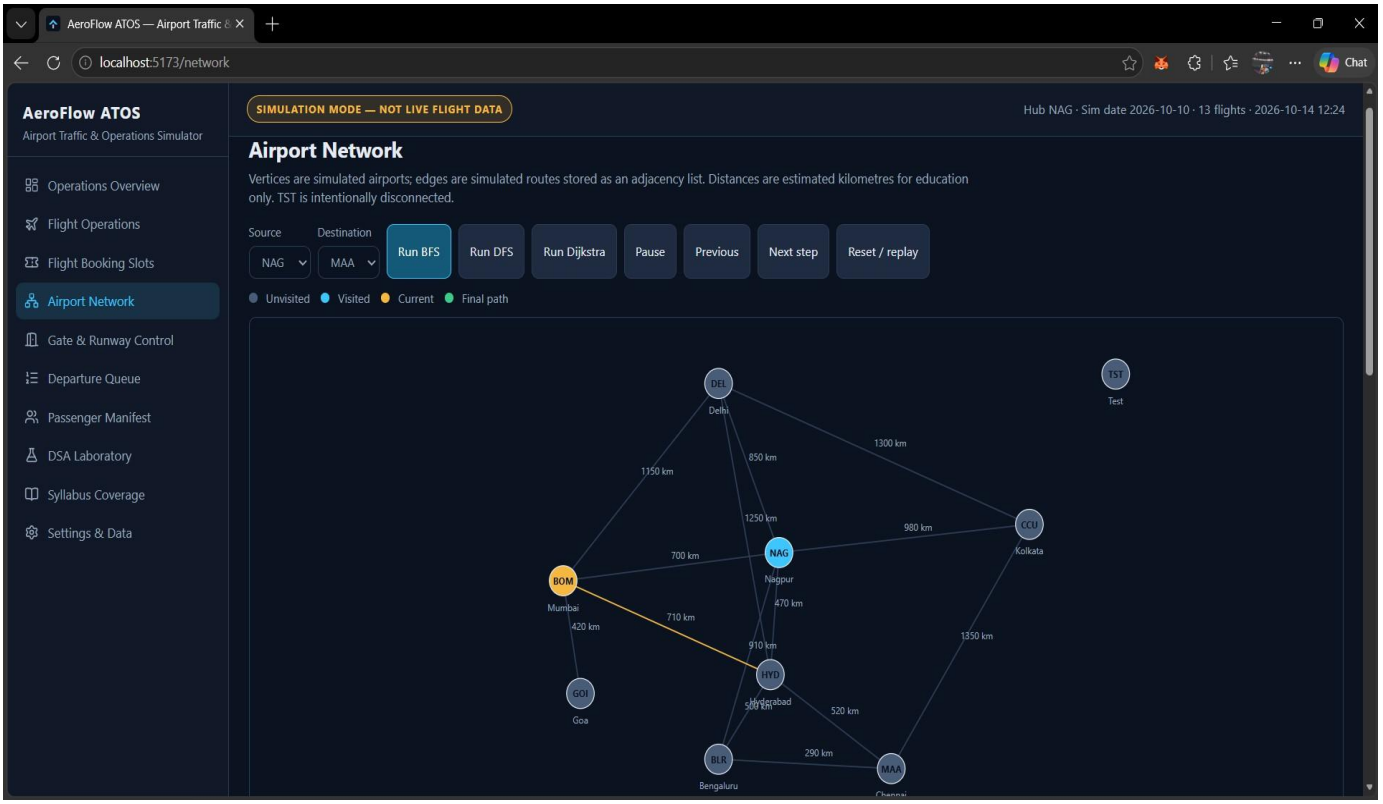
In conclusion, AeroFlow ATOS serves as an educational platform for understanding how data structures and algorithms can support flight management and airport operations. The project can be extended in the future by introducing database integration, user authentication, advanced analytics, and additional airport-management features. The current system is a simulation intended for academic learning rather than real-world airport operations.

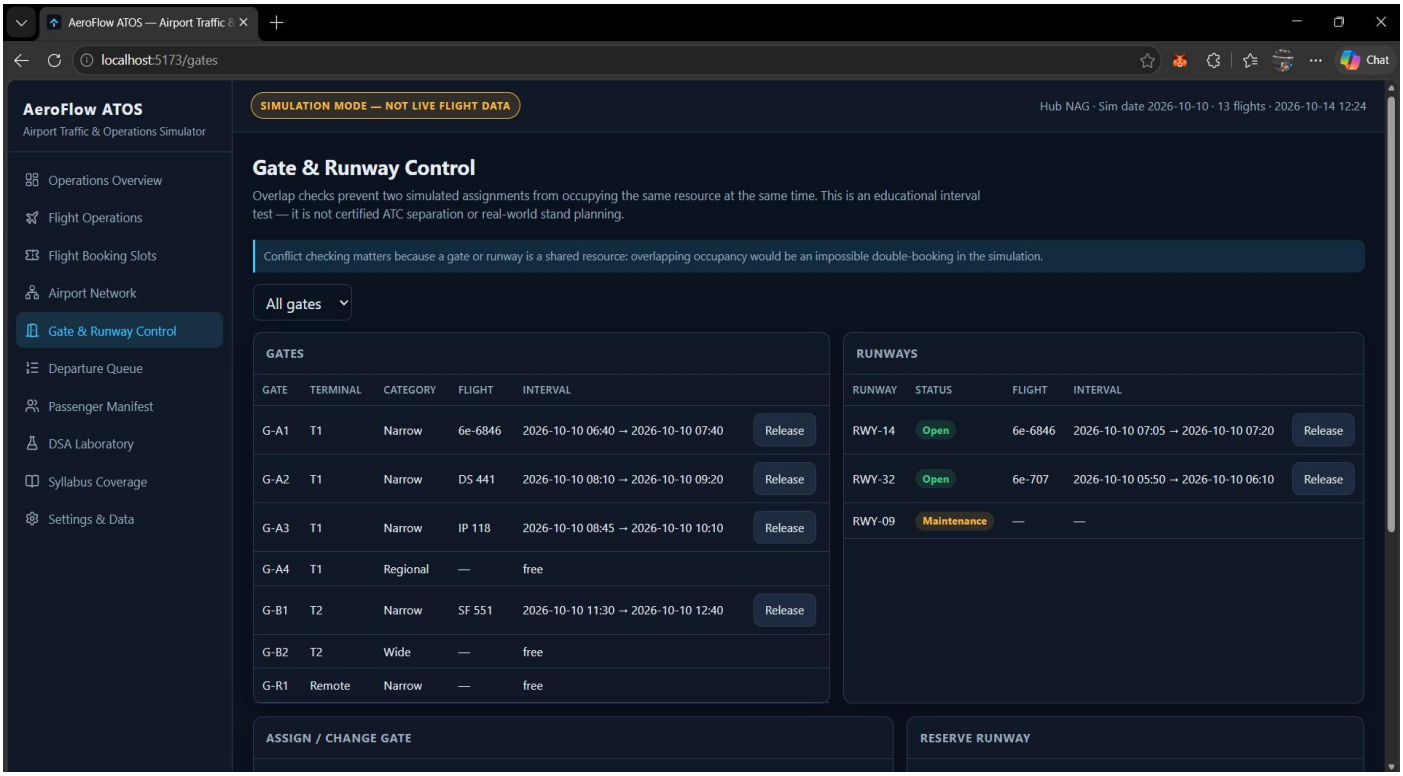
7. Outputs

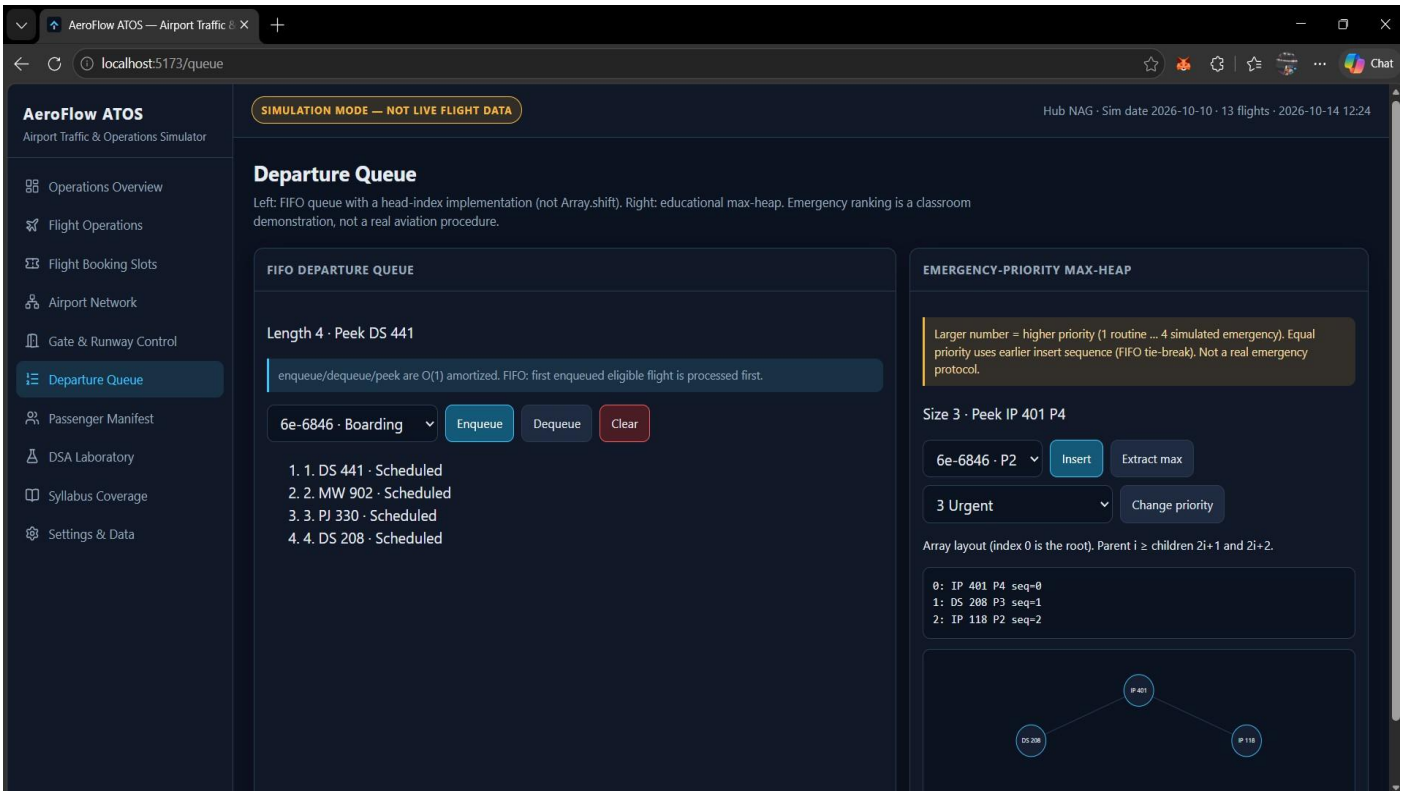

localhost:5173
AeroFlow ATOS
Airport Traffic & Operations Simulator
SIMULATION MODE — NOT LIVE FLIGHT DATA
Hub NAG · Sim date 2026-10-10 · 13 flights · 2026-10-14 12:24
Operations Overview
Flight Operations
Flight Booking Slots
Airport Network
Gate & Runway Control
Departure Queue
Passenger Manifest
DSA Laboratory
Syllabus Coverage
Settings & Data
Operations Overview
Simulated control picture for hub NAG. Totals are derived from current records — not hardcoded. All times follow the selected simulation date 2026-10-10.
Restore demonstration data
Undo last action
TOTAL FLIGHTS 13
SCHEDULED DEPARTURES 5 from NAG
SCHEDULED ARRIVALS 5 into NAG
DELAYED FLIGHTS 2
CANCELLED FLIGHTS 1
AVAILABLE GATES 3
OCCUPIED GATES 4
SIMULATED OTP 83% non-cancelled, delay = 0
UPCOMING DEPARTURES
FLIGHT AIRLINE DEST TIME GATE STATUS
6e-6846 IndiGo DEL 07:10 G-A1 Boarding
IP 118 IndiGo DEL 09:15 G-A3 Delayed
PJ 330 Qantas HYD 11:20 — Scheduled
DS 208 Singapore Airlines MAA 13:30 — Scheduled
ARRIVALS OVERVIEW
FLIGHT AIRLINE ORIGIN TIME GATE STATUS
6e-707 Singapore Airlines BOM 08:20 G-B2 Departed
DS 441 Air India BOM 10:15 G-A2 Scheduled
MW 902 Vistara DEL 12:00 — Scheduled
SF 551 Qatar Airways BLR 13:40 G-B1 Scheduled

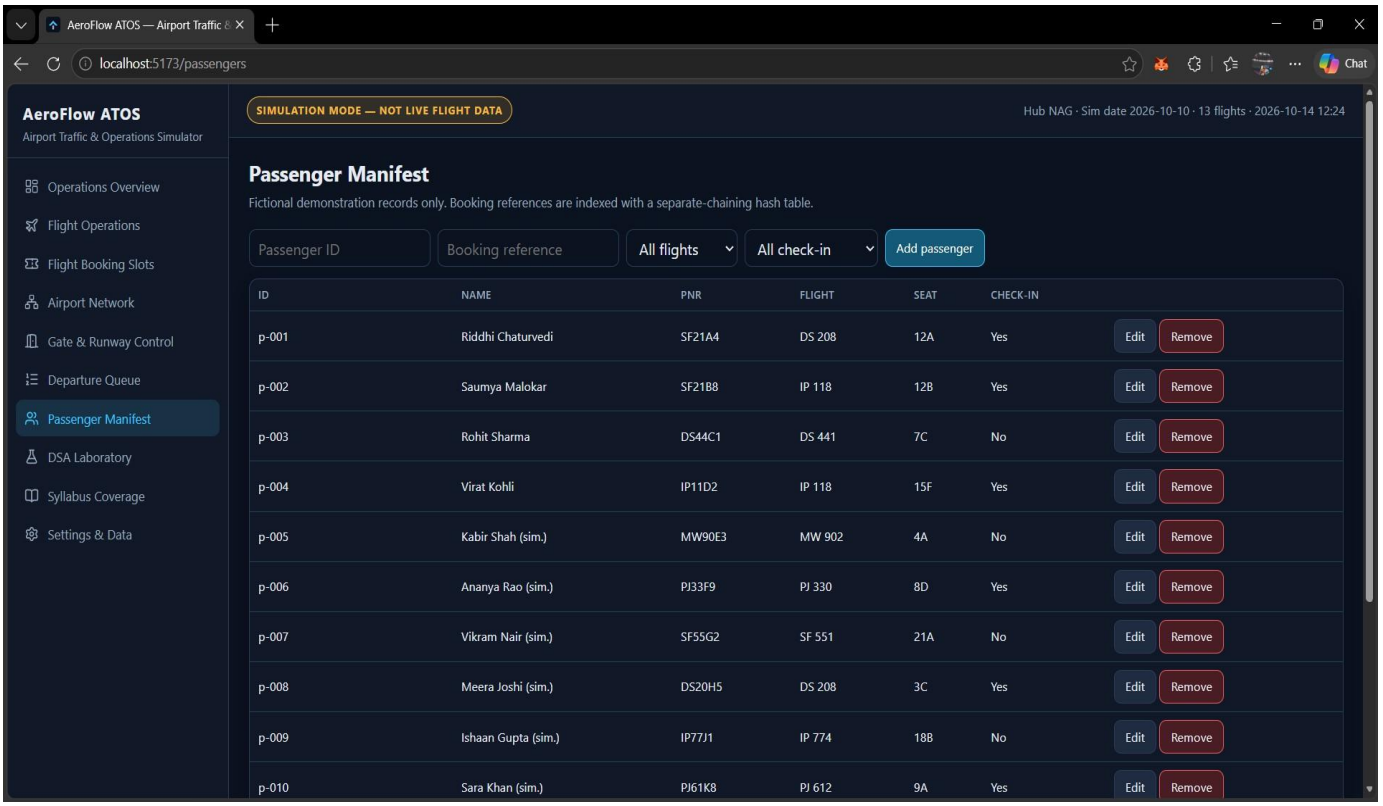

AeroFlow ATOS — Airport Traffic &
localhost:5173/flights
Chat
AeroFlow ATOS
Airport Traffic & Operations Simulator
SIMULATION MODE — NOT LIVE FLIGHT DATA
Hub NAG · Sim date 2026-10-10 · 13 flights · 2026-10-14 12:24
Operations Overview
Flight Operations
Flight Booking Slots
Airport Network
Gate & Runway Control
Departure Queue
Passenger Manifest
DSA Laboratory
Syllabus Coverage
Settings & Data
Flight Operations
All sample schedules, routes, flight numbers, assignments, and status changes are educational demonstration data. They are not live airline records.
Search flight number
Search airline
All statuses
Origin
Destination
All flows
Sort: departure
Add flight
FLIGHT AIRLINE ROUTE STD GATE STATUS DELAY Q POS
AI 101 Air India NAG → BOM 2026-10-10 05:30 G-A3 Arrived 0m — Edit Delete
6e-707 Singapore Airlines BOM → NAG 2026-10-10 06:00 G-B2 Departed 0m — Edit Delete
6e-6846 IndiGo NAG → DEL 2026-10-10 07:10 G-A1 Boarding 0m — Edit Delete
DS 441 Air India BOM → NAG 2026-10-10 08:40 G-A2 Scheduled 0m 1 Edit Delete
IP 118 IndiGo NAG → DEL 2026-10-10 09:15 G-A3 Delayed 35m — Edit Delete
MW 902 Vistara DEL → NAG 2026-10-10 10:05 — Scheduled 0m 2 Edit Delete
PJ 330 Qantas NAG → HYD 2026-10-10 11:20 — Scheduled 0m 3 Edit Delete
SF 551 Qatar Airways BLR → NAG 2026-10-10 12:00 G-B1 Scheduled 0m — Edit Delete


AeroFlow ATOS — Airport Traffic &
localhost:5173/booking
Chat
AeroFlow ATOS
Airport Traffic & Operations Simulator
Operations Overview
Flight Operations
Flight Booking Slots
Airport Network
Gate & Runway Control
Departure Queue
Passenger Manifest
DSA Laboratory
Syllabus Coverage
Settings & Data
SIMULATION MODE — NOT LIVE FLIGHT DATA
Hub NAG · Sim date 2026-10-10 · 13 flights · 2026-10-14 12:24
Flight Booking Slots
Reserve seats on scheduled flights across IndiGo, Air India, Singapore Airlines, Emirates, Qatar Airways, Qantas, and Vistara. Each reservation updates passenger counts, syncs with the separate-chaining Hash Table, and generates an educational boarding pass.
10 Active Booking Slots Available
All Airlines (IndiGo, Air India, Emirates, etc.)
Any Origin
Any Destination
All Cabin Classes
IndiGo
Air India
Singapore Airlines
Emirates
Qatar Airways
Qantas
Vistara
IndiGo
Flight 6e-6846 · None (Codeshare)
Boarding
NAG
STD 07:10
Direct Flight
Airbus A320
DEL
STA 08:40
Seat Availability Slot:
32 seats available (148/180)
Economy Slot
₹3,800 / seat
Business Slot
₹10,640 / seat
Terminal T1
Gate G-A1
Narrow Body
Instant PNR Generation
Book This Slot
Air India
Flight DS 441 · Star Alliance
Scheduled
BOM
STD 08:40
Direct Flight
Airbus A321
NAG
STA 10:15
Seat Availability Slot:
56 seats available (164/220)
Economy Slot
₹3,800 / seat
Business Slot
₹10,640 / seat
Terminal T1
Gate G-A2
Narrow Body
Instant PNR Generation
Book This Slot


AeroFlow ATOS — Airport Traffic &
localhost:5173/network
Chat
AeroFlow ATOS
Airport Traffic & Operations Simulator
Operations Overview
Flight Operations
Flight Booking Slots
Airport Network
Gate & Runway Control
Departure Queue
Passenger Manifest
DSA Laboratory
Syllabus Coverage
Settings & Data
SIMULATION MODE — NOT LIVE FLIGHT DATA
Hub NAG · Sim date 2026-10-10 · 13 flights · 2026-10-14 12:24
Airport Network
Vertices are simulated airports; edges are simulated routes stored as an adjacency list. Distances are estimated kilometres for education only. TST is intentionally disconnected.
Source
NAG
Destination
MAA
Run BFS
Run DFS
Run Dijkstra
Pause
Previous
Next step
Reset / replay
Unvisited
Visited
Current
Final path
DEL
Delhi
TST
Test
BOM
Mumbai
NAG
Nagpur
CCU
Kolkata
HYD
Hyderabad
GOI
Goa
BLR
Bengaluru
MAA
Chennai
1300 km
850 km
1150 km
1250 km
980 km
700 km
470 km
710 km
420 km
910 km
1350 km
520 km
290 km


AeroFlow ATOS — Airport Traffic &
localhost:5173/gates
Chat
AeroFlow ATOS
Airport Traffic & Operations Simulator
Operations Overview
Flight Operations
Flight Booking Slots
Airport Network
Gate & Runway Control
Departure Queue
Passenger Manifest
DSA Laboratory
Syllabus Coverage
Settings & Data
SIMULATION MODE — NOT LIVE FLIGHT DATA
Hub NAG · Sim date 2026-10-10 · 13 flights · 2026-10-14 12:24
Gate & Runway Control
Overlap checks prevent two simulated assignments from occupying the same resource at the same time. This is an educational interval test — it is not certified ATC separation or real-world stand planning.
Conflict checking matters because a gate or runway is a shared resource: overlapping occupancy would be an impossible double-booking in the simulation.
All gates
GATES
GATE TERMINAL CATEGORY FLIGHT INTERVAL
G-A1 T1 Narrow 6e-6846 2026-10-10 06:40 → 2026-10-10 07:40 Release
G-A2 T1 Narrow DS 441 2026-10-10 08:10 → 2026-10-10 09:20 Release
G-A3 T1 Narrow IP 118 2026-10-10 08:45 → 2026-10-10 10:10 Release
G-A4 T1 Regional — free
G-B1 T2 Narrow SF 551 2026-10-10 11:30 → 2026-10-10 12:40 Release
G-B2 T2 Wide — free
G-R1 Remote Narrow — free
RUNWAYS
RUNWAY STATUS FLIGHT INTERVAL
RWY-14 Open 6e-6846 2026-10-10 07:05 → 2026-10-10 07:20 Release
RWY-32 Open 6e-707 2026-10-10 05:50 → 2026-10-10 06:10 Release
RWY-09 Maintenance — —
ASSIGN / CHANGE GATE
RESERVE RUNWAY


AeroFlow ATOS — Airport Traffic &
localhost:5173/queue
Chat
AeroFlow ATOS
Airport Traffic & Operations Simulator
Operations Overview
Flight Operations
Flight Booking Slots
Airport Network
Gate & Runway Control
Departure Queue
Passenger Manifest
DSA Laboratory
Syllabus Coverage
Settings & Data
SIMULATION MODE — NOT LIVE FLIGHT DATA
Hub NAG · Sim date 2026-10-10 · 13 flights · 2026-10-14 12:24
Departure Queue
Left: FIFO queue with a head-index implementation (not Array.shift). Right: educational max-heap. Emergency ranking is a classroom demonstration, not a real aviation procedure.
FIFO DEPARTURE QUEUE
Length 4 · Peek DS 441
enqueue/dequeue/peek are O(1) amortized. FIFO: first enqueued eligible flight is processed first.
6e-6846 · Boarding
Enqueue
Dequeue
Clear
1. 1. DS 441 · Scheduled
2. 2. MW 902 · Scheduled
3. 3. PJ 330 · Scheduled
4. 4. DS 208 · Scheduled
EMERGENCY-PRIORITY MAX-HEAP
Larger number = higher priority (1 routine … 4 simulated emergency). Equal priority uses earlier insert sequence (FIFO tie-break). Not a real emergency protocol.
Size 3 · Peek IP 401 P4
6e-6846 · P2
Insert
Extract max
3 Urgent
Change priority
Array layout (index 0 is the root). Parent i ≥ children 2i+1 and 2i+2.
0: IP 401 P4 seq=0
1: DS 208 P3 seq=1
2: IP 118 P2 seq=2
IP 401
DS 208
IP 118


AeroFlow ATOS — Airport Traffic &
localhost:5173/passengers
Chat
AeroFlow ATOS
Airport Traffic & Operations Simulator
Operations Overview
Flight Operations
Flight Booking Slots
Airport Network
Gate & Runway Control
Departure Queue
Passenger Manifest
DSA Laboratory
Syllabus Coverage
Settings & Data
SIMULATION MODE — NOT LIVE FLIGHT DATA
Hub NAG · Sim date 2026-10-10 · 13 flights · 2026-10-14 12:24
Passenger Manifest
Fictional demonstration records only. Booking references are indexed with a separate-chaining hash table.
Passenger ID
Booking reference
All flights
All check-in
Add passenger
ID NAME PNR FLIGHT SEAT CHECK-IN
p-001 Riddhi Chaturvedi SF21A4 DS 208 12A Yes Edit Remove
p-002 Saumya Malokar SF21B8 IP 118 12B Yes Edit Remove
p-003 Rohit Sharma DS44C1 DS 441 7C No Edit Remove
p-004 Virat Kohli IP11D2 IP 118 15F Yes Edit Remove
p-005 Kabir Shah (sim.) MW90E3 MW 902 4A No Edit Remove
p-006 Ananya Rao (sim.) PJ33F9 PJ 330 8D Yes Edit Remove
p-007 Vikram Nair (sim.) SF55G2 SF 551 21A No Edit Remove
p-008 Meera Joshi (sim.) DS20H5 DS 208 3C Yes Edit Remove
p-009 Ishaan Gupta (sim.) IP77J1 IP 774 18B No Edit Remove
p-010 Sara Khan (sim.) PJ61K8 PJ 612 9A Yes Edit Remove